

AGENTIC AI WITH LANGGRAPH – SMART CAREER GUIDANCE AGENT

Objective

The objective of this project is to develop an Agentic AI based Smart Career Guidance System using LangGraph. Multiple AI agents collaborate to analyze resumes, identify skill gaps, recommend suitable career paths, generate personalized learning roadmaps, and provide relevant job suggestions.

Algorithm

1. Install required libraries.
2. Configure the .env file and load the OpenAI/Gemini API key.
3. Load career datasets from Kaggle or Hugging Face.
4. Initialize LangGraph workflow and create Resume Analysis, Skill Gap Detection, Career Recommendation, Learning Roadmap, Job Matching and Report Generation agents.
5. Accept resume and user details.
6. Analyze the resume and extract skills, education and certifications.
7. Compare skills with the dataset to identify missing skills.
8. Use the LLM to generate career recommendations.
9. Generate a personalized learning roadmap.
10. Match the profile with available job roles.
11. Generate the final career guidance report.

Inputs

Resume (PDF/DOCX), Educational Details, Technical Skills, Certifications, Projects and Career Interests.

Outputs

Resume Analysis Report, Skill Gap Analysis, Career Recommendations, Personalized Learning Roadmap, Job Suggestions and Final Career Guidance Report.

Conditions

If the resume is not uploaded, display an error message. If missing skills are identified, generate a learning roadmap; otherwise recommend suitable jobs directly.

Loops

For each extracted skill, compare it with industry skills. For each available job role, calculate the profile matching score and rank the jobs.

Logic

User Input → Resume Analysis → Skill Gap Detection → Career Recommendation → Learning Roadmap → Job Matching → Final Report.

Required Libraries

LangGraph, LangChain, OpenAI API / Gemini API, FAISS, Pandas, NumPy, Streamlit, python-dotenv.

Dataset

Career Skills Dataset, Job Roles Dataset, Kaggle Career Dataset, Hugging Face Dataset and LinkedIn Skills Data.

Model / LLM

OpenAI GPT or Google Gemini.

Expected Result

The system provides intelligent career guidance through multiple AI agents coordinated using LangGraph. Users receive personalized career recommendations, learning plans, job suggestions and a comprehensive career guidance report.